

SYNTHETIC DATA FOR TEACHING — NOT A REAL STUDY. Patients, investigators and institution are fictional.

This demonstration manuscript was written by AI (Claude Opus 5.5). In a real submission, the text must be written by the authors.

Response to the Editor and Reviewers

We thank the editor and the two reviewers for their careful reading and constructive comments, which have improved the manuscript. We have addressed every comment below. In the marked revised manuscript, new text is shown in blue and underlined and deleted text in red and struck through; the Summary of Changes below gives the page of each change. Analyses added in response to the reviewers were not prespecified and are labeled post hoc throughout the manuscript.

Summary of Changes

Page numbers refer to the marked revised manuscript. New text is shown in blue and underlined; deleted text in red and struck through.

Section	Part	Page(s)
Abstract	—	8–9
Materials and Methods	Treatment Groups and Outcomes	11
Materials and Methods	Statistical Analysis	12–13
Results	Patients	14
Results	Event-Free and Overall Survival	15
Results	Post Hoc Analyses	16

Section	Part	Page(s)
Discussion	—	16–19
References	Added: Schneeweiss et al., 2013 (reference 16)	—
Supplementary material	Supplementary Table S4 (post hoc analyses)	—
Supplementary material	Supplementary Figure S2 (propensity-score distributions)	—

Editorial Office

Comment E1. *The structured abstract must not exceed 250 words and must use the headings Background, Materials and Methods, Results, Conclusion.*

Response: The revised structured abstract remains within 250 words and uses the four required headings: Background, Materials and Methods, Results, and Conclusion.

Changes: Abstract (revised to report the post hoc worst-case analysis and a tempered conclusion).

Comment E2. *In-text citations must be Arabic numerals in superscript square brackets placed after punctuation, and the reference list must follow the journal (Vancouver/ICMJE) style.*

Response: In-text citations are Arabic numerals in superscript square brackets placed after punctuation, and the reference list follows the journal’s Vancouver style, with the first six authors followed by et al. and journal abbreviations according to Index Medicus. Every reference was verified against PubMed and Crossref.

Changes: References (one reference added for the TRYPHAENA trial; numbering updated).

Comment E3. *Figure legends must be 40 words or fewer. Please also confirm the Data Availability Statement and the disclosure of any generative AI use required by the journal.*

Response: All figure legends contain 40 words or fewer. The Data Availability Statement and the Use of Generative AI statement appear before the References.

Changes: No change needed.

Reviewer 1

Comment R1.1. *Treatment changed with calendar time. Dual blockade was used mostly in the later years and single blockade mostly in the earlier years. Please explain how the analysis separates the effect of the regimen from the effect of the era, and show whether the two groups truly overlap (for example, propensity score distributions by group).*

Response: We agree that separating the effect of the regimen from that of the treatment era is the central challenge of this study, and we addressed it in three ways. First, the year of treatment start was included in the propensity-score model, and overlap weighting was chosen because the two eras overlapped little; this method gives the greatest weight to patients who could plausibly have received either regimen and avoids extreme weights. Second, as suggested, we now show the propensity-score distributions by group (new Supplementary Figure S2). The overlap was partial: the median propensity score was 0.99 with dual blockade and 0.05 with trastuzumab alone, and patients who started treatment in 2016–2019 contributed 67.6% of the overlap weights. Third, the prespecified analysis restricted to 2016–2019 gave an odds ratio (OR) of 2.15 (95% CI, 0.92–5.01), and a new post hoc analysis that modeled year with a natural spline gave an OR of 2.00 (95% CI, 1.05–3.79). We state in the Discussion that the estimate applies to patients for whom both regimens were realistic options.

Changes: Methods (Statistical Analysis, post hoc analyses); Results (Patients; Post Hoc Analyses); Discussion (third paragraph); Supplementary Figure S2 and Table S4.

Comment R1.2. *Adjuvant T-DM1 for residual disease became available during the later part of the study period. Because this was given mainly in the dual-blockade era, it may account for part of the EFS difference. Please address this with a sensitivity analysis.*

Response: We agree that adjuvant trastuzumab emtansine (T-DM1) was used mainly in the dual-blockade era: among patients without pathologic complete response (pCR), 59.7% in the dual-blockade group and 4.6% in the single-blockade group received T-DM1. We did not stratify the event-free survival (EFS) analysis by T-DM1, because T-DM1 is given only to patients with residual disease and therefore depends on the response to the neoadjuvant regimen; conditioning on it would introduce bias. Instead, we performed a post hoc analysis restricted to patients who started treatment in 2012–2019, before adjuvant T-DM1 was used at the study hospital. In this group, none of the patients without pCR received T-DM1, and the hazard ratio (HR) for EFS was 0.81 (95% CI, 0.39–1.70), similar to that in the full cohort (HR, 0.89 [95% CI, 0.47–1.69]). We also note that the primary analysis did not show a clear EFS difference between the regimens, and we have kept the conclusion that the data are insufficient to assess EFS.

Changes: Methods (Statistical Analysis, post hoc analyses); Results (Post Hoc Analyses); Discussion (fourth paragraph); Supplementary Table S4.

Comment R1.3. *Patients who progressed before surgery or had surgery elsewhere have no pathologic response. Please state how they were handled in the pCR analysis and whether the conclusion changes under alternative handling.*

Response: As stated in the Methods, patients whose disease progressed before surgery were classified as not achieving pCR. Excluding these patients (four with dual blockade and five with trastuzumab alone) gave an OR of 2.01 (95% CI, 1.12–3.62). Patients operated on at another

hospital were excluded because no pathology report was available. In new post hoc best-case and worst-case analyses that included these six dual-blockade and ten single-blockade patients, the OR was 2.19 (95% CI, 1.27–3.77) if none of them achieved pCR and 1.40 (95% CI, 0.77–2.53) if all did. Because most of these patients received trastuzumab alone, the worst-case assumption attenuated the association, and its confidence interval included the null value. We report this result in the Abstract, Results, and limitations and have tempered the Conclusion accordingly.

Changes: Abstract; Methods (Statistical Analysis, post hoc analyses); Results (Post Hoc Analyses); Discussion (limitations and conclusion); Supplementary Table S4.

Comment R1.4. *Residual confounding is likely in a retrospective cohort. Please quantify how strong an unmeasured confounder would need to be to explain away the main findings.*

Response: We agree that residual confounding is likely. The E-value for the primary OR was 2.19, and that for its lower confidence limit was 1.31; an unmeasured confounder associated with both treatment and pCR by a risk ratio of approximately 2.19, beyond the measured covariates, could explain away the observed association. These values are reported in the Results and discussed in the limitations, where we have corrected the scale of the E-value to the risk ratio.

Changes: Results (Sensitivity and Subgroup Analyses); Discussion (limitations); Supplementary Table S3.

Reviewer 2

Comment R2.1. *Please define time zero precisely and explain how the design avoids immortal time bias.*

Response: Time zero was the index date, defined as the date of the first neoadjuvant anticancer drug order, when eligibility was met and the treatment strategy began; all outcomes were measured from this date. Because treatment groups were defined by any pertuzumab use before

surgery, a patient could in principle be classified as receiving single blockade because surgery or progression occurred before pertuzumab was added. We examined this post hoc. Pertuzumab was started on the index date in 300 patients and later in 116 patients (median, day 84, consistent with sequential regimens), whereas the earliest surgery or disease progression in the single-blockade group occurred on day 97. Therefore, no patient was assigned to single blockade because of an early event, and immortal time bias is unlikely.

Changes: Methods (Treatment Groups and Outcomes); Results (Post Hoc Analyses).

Comment R2.2. *Follow-up is shorter for patients treated in recent years. Please report median follow-up by group (reverse Kaplan–Meier) and consider a restricted time horizon for the survival comparisons.*

Response: Median follow-up was estimated with the reverse Kaplan-Meier method and reported by group (5.1 years for dual blockade and 10.7 years for trastuzumab alone); the method is now stated in the Methods. As suggested, we also report a restricted time horizon: the five-year restricted mean survival time difference was 0.17 years (95% CI, −0.39 to 0.73) for EFS and 0.08 years (95% CI, −0.30 to 0.47) for overall survival (OS), and with follow-up truncated at five years the HRs were 0.59 (95% CI, 0.26–1.36) for EFS and 0.77 (95% CI, 0.30–1.99) for OS (post hoc). These results do not change our conclusions.

Changes: Methods (Statistical Analysis); Results (Event-Free and Overall Survival; Post Hoc Analyses); Supplementary Table S4.

Comment R2.3. *The overall survival analysis has few events. The conclusion regarding OS should be worded cautiously, with the confidence interval emphasized rather than the P value.*

Response: We agree. The OS analysis was exploratory and had few events. We now emphasize the confidence intervals rather than the P values, state that the wide confidence interval for OS is compatible with both a clinically important benefit and harm, and state in the Discussion that the

OS results should be regarded as exploratory. The Conclusion states that the data were insufficient to determine the effect on OS.

Changes: Results (Event-Free and Overall Survival); Discussion (fourth paragraph).

Comment R2.4. *The Discussion should compare the findings with NeoSphere, TRYPHAENA, and published real-world cohorts, and should list the limitations of a single-center design.*

Response: We now compare our findings with NeoSphere, TRYPHAENA, and PEONY, with published real-world cohorts from Turkey, Italy, China, and Spain, and with a meta-analysis of real-world studies. We have also expanded the first limitation to describe the consequences of the single-center design: the results reflect one institution's patient population, treatment practices, and pathology procedures and may not apply to other settings.

Changes: Discussion (second paragraph and limitations); References.

We hope that the revised manuscript is now suitable for publication in the *Journal of Cancer Research and Practice*.